

Urban Tree Cooling v6.2 - Pre-Gate A Decision Summary

Updated 3 September 2026 | Outcome-blind free checks and nonthermal acquisition closeout | No new v6.2 coefficient viewed

DECISION POSTURE: Do not begin Gate A under the current frozen design. Keep the optional atmospheric-demand branch inactive and on hold pending a supervisor ruling.

What passed

Foundation and catalogue. The independent Collection 3 audit passed all 31 required checks. Offline reproduction succeeded; all 69 evidence-packet checksums and 12 input hashes matched. Historical counts remain provenance only, not the new study sample.

Connectivity for planning. All eight Phoenix/Los Angeles combinations of two season windows and 15/25 degree view sets were reported, including block-pass coverage, component share, sectors and land-use dominance. This supports planning, subject to the historical-proxy limitations below.

Data verification. The inventory contains 149 HLS Fmask, 14 official Science TCC and two 3DEP files: 165 files and 800,091,676 bytes. Every file has a local SHA-256; the 16 Drive TCC/3DEP files also match Google MD5 metadata.

Checks executed correctly. The official TCC screen and Fmask lead-lag reassessment completed without opening HLS reflectance, ECOSTRESS LST or new coefficients. Exact HLS dates and acquisition IDs are now verified, and 41 distinct cloud-screened pre/post pairs were found.

What remains unresolved

Design choice. The supervisor must choose whether to stop, reframe the estimand, or approve a prospective design amendment after the canopy-span failure. The observed result cannot be used to relax the threshold retroactively.

Deferred inputs. Full HLS reflectance, ECOSTRESS Collection 3 thermal products and precipitation are not yet acquired for the final analysis. HRRR/weather, L1B geometry, land use and other context are partial or inherited and must be frozen against the final sample if work restarts.

Supervisor confirmations. A keep/drop/amend ruling is still needed for atmospheric demand, and the single email reference to v6.1 should be confirmed as a typo; v6.2 remains controlling because the guide, proposal, schedule and branch conventions all specify v6.2.

Whether the optional atmospheric-demand branch should remain

Recommendation: retain only as an inactive, documented option. It should not enter a confirmatory model now. The binding full-nuisance test fails both cities. A less restrictive sensitivity passes Phoenix, while Los Angeles misses the common-support-width floor (0.300 versus 0.500 kPa). This mixed evidence supports supervisor review, not activation.

If the supervisor wants to retain the scientific question, the revised nuisance design and keep/drop rule must be approved and frozen prospectively before any new coefficient is viewed. Otherwise, follow the binding rule and drop both VPD interactions together.

What failed

Stage 1 canopy support - controlling failure. Across the five preselected passes, zero of 13,509 candidate block-passes met the frozen Science TCC p10-p90 canopy-span floor of 0.20. The median span was 0.06600 and the maximum was 0.18410. Because this was an optimistic nonthermal screen, adding thermal-complete masking cannot create an eligible block-pass; the conditional thermal Stage 1 rerun was therefore not performed.

Lead-lag confirmatory support. Fmask screening verified 41 of 110 candidate pass-window-sensor pairs (37.3%): Phoenix 20/44 and Los Angeles 21/66. Sixty-three candidates had timing-valid images but fewer than 30 shared cloud-free blocks, and six lacked a timing-valid pair in the downloaded plan. Neither city had a season window reaching the frozen 70% requirement in every applicable year-sensor stratum; Los Angeles also failed acquisition reuse in three strata. Lead-lag remains exploratory.

Atmospheric-demand binding test. The predeclared strict VPD support rule fails both cities after the full temperature, vapour-pressure, time, season and viewing-geometry adjustment. The later parsimonious analysis is explicitly nonbinding and does not reverse that result.

Whether Gate A should begin

No - not under the current frozen design. Foundation and connectivity readiness do not overcome the absence of an estimable canopy contrast. With zero eligible block-passes under the required official canopy product, the planned within-block canopy-LST slope cannot enter the requested Stage 1 precision census.

Gate A can be reconsidered only after supervisor review produces either a defensible reframing or a prospectively frozen amendment. That decision must be logged before acquiring or opening Gate A-only reflectance and thermal outcome data, and the amended design must rerun the affected free checks.

Any deviations from the Working Guide

Version discrepancy. One email reference says v6.1; all controlling attachments and repository conventions say v6.2, which is being used pending correction.

Historical proxies. Connectivity uses inherited Collection 2 incidence rather than a selected Collection 3 v6.2 sample. The first Stage 1 diagnostic used modified NLCD canopy, Collection 2 thermal data, LSTE height and other inherited context proxies; it is superseded and is not the controlling scientific result.

Lead-lag scope. The reassessment uses Fmask from 149 catalogue-selected acquisitions for feasibility only. It does not construct the Working Guide's final one-sided antecedent exposure or future placebo from raw HLS reflectance, and un-downloaded alternative HLS acquisitions could improve failed rows.

Pending rather than final VPD ruling. The Working Guide requests a definite keep/drop decision. At the user's direction, the branch is currently held inactive pending supervisor review; this is explicitly disclosed and no interaction may be fitted meanwhile.

No frozen scientific threshold was silently relaxed. Prior-protocol and synthetic outputs remain archived or isolated, and no new thermal value, reflectance value or coefficient was opened for these free checks.

Decision basis: protocol v6.2; decisions V6.2-D003/D005/D006-D019; independent foundation audit; D1a-D1d evidence packages; and the verified raw-asset inventory. This is a decision aid, not supervisor approval.